

SINGAPORE SELETAR, Singapore

WMO: 486990

Lat: 1.417N

Lon: 103.868E

Elev: 11

StdP: 101.19

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	23.9	19.0	13.8	31.5	19.9	14.6	30.6	9.1	29.1	8.4	29.2	2.4	220	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.2	34.2	25.1	33.9	25.1	33.1	25.1	27.0	30.2	26.7	29.9	26.6	29.8	3.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	28.1	26.1	21.5	28.1	25.9	21.2	28.0	85.1	30.8	83.9	30.3	83.0	29.9	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.6	5.9	22.7	35.5	0.5	1.2	22.3	36.4	22.0	37.1	21.8	37.8	21.4	38.7		
(5)			20.9	27.7	0.5	0.4	20.6	27.9	20.3	28.1	20.0	28.3	19.6	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	27.7	28.3	28.8	29.1	29.2	29.1	28.8	28.8	28.7	28.7	28.2	27.9		
	DBStd	1.25	1.21	0.98	1.15	1.09	1.09	1.35	1.33	1.32	1.18	1.11	0.97	1.15		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6789	549	513	581	572	596	572	583	582	561	580	547	553		
	CDD18.3	3748	291	280	323	322	338	322	325	323	311	322	297	295		
	CDH23.3	40818	2780	2822	3422	3555	3959	3835	3794	3743	3517	3544	2987	2860		
	CDH26.7	14978	821	953	1238	1364	1617	1576	1506	1461	1356	1324	942	820		
(14) Wind	WSAvg	2.5	3.7	4.0	3.0	2.0	1.9	2.0	2.4	2.5	2.2	2.0	2.0	2.8		
(15) Precipitation	PrecAvg	2237	184	173	197	194	183	170	151	115	175	175	231	289		
	PrecMax	2968	457	421	389	351	341	274	247	201	512	368	414	548		
	PrecMin	1686	25	3	43	39	91	74	20	32	60	45	71	78		
	PrecStd	387	106	105	102	81	71	56	65	51	113	91	101	131		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	34.0	34.2	34.9	34.8	34.9	34.1	34.1	34.0	34.2	33.2	33.2		
		MCWB	24.6	24.6	24.8	25.5	25.9	24.9	25.2	24.7	24.8	24.6	25.3	25.0		
	2%	DB	32.1	33.0	33.2	33.9	33.8	33.9	33.1	33.1	33.1	33.2	32.8	32.1		
		MCWB	24.7	24.6	24.8	25.5	25.7	25.2	25.2	25.0	24.9	24.7	25.2	25.0		
	5%	DB	31.2	32.1	32.8	33.0	33.0	33.0	32.2	32.2	32.2	32.8	31.9	31.2		
		MCWB	24.8	24.5	24.8	25.5	25.7	25.4	25.1	24.9	24.9	24.8	25.2	25.1		
	10%	DB	30.8	31.1	31.9	32.1	32.1	32.1	31.9	31.8	31.9	31.9	31.0	30.2		
		MCWB	24.8	24.5	24.9	25.4	25.7	25.4	25.1	25.0	24.9	25.0	25.4	25.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.2	26.7	27.1	27.4	27.2	26.7	26.7	26.7	26.9	26.7	26.7		
		MCDB	29.4	29.7	30.5	30.5	30.7	30.2	29.8	29.8	29.8	29.6	29.6	29.4		
	2%	WB	26.1	25.7	26.2	26.7	27.0	26.7	26.5	26.3	26.3	26.5	26.4	26.2		
		MCDB	28.7	29.4	29.7	30.0	30.4	30.0	29.7	29.6	29.5	29.4	29.3	28.9		
	5%	WB	25.7	25.5	25.9	26.5	26.7	26.5	26.2	26.1	26.1	26.2	26.1	26.0		
		MCDB	28.5	29.1	29.3	29.8	30.0	29.8	29.5	29.5	29.3	29.2	28.9	28.6		
	10%	WB	25.4	25.1	25.6	26.2	26.5	26.2	26.0	25.7	25.7	25.9	25.9	25.7		
		MCDB	28.4	28.8	29.0	29.5	29.8	29.5	29.4	29.2	29.1	29.0	28.6	28.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.8	7.0	7.1	6.6	6.2	6.1	6.1	6.4	6.7	6.6	5.9		
		MCDBR	6.9	7.7	7.9	7.8	7.3	7.0	6.6	6.6	6.9	7.4	7.5	7.0		
	5% WB	MCWBR	2.2	2.0	2.1	2.4	2.4	2.4	2.2	2.2	2.3	2.4	2.3	2.3		
		MCWBR	6.1	6.8	7.0	7.1	6.7	6.3	6.3	6.1	6.5	6.7	6.7	6.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.470	0.483	0.493	0.492	0.478	0.490	0.482	0.524	0.555	0.581	0.485	0.470		
		taud	2.293	2.239	2.219	2.223	2.269	2.239	2.270	2.129	2.039	1.946	2.249	2.300		
	Ebn at Noon	Ebn at Noon	861	855	842	824	812	790	803	787	779	763	841	854		
		Edh at Noon	137	147	150	145	134	136	133	158	177	195	142	135		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.54	5.25	5.23	5.07	4.69	4.45	4.42	4.48	4.47	4.48	4.25	4.14		
	RadStd	RadStd	0.38	0.47	0.46	0.32	0.22	0.32	0.33	0.25	0.31	0.33	0.26	0.31		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.37	+0.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page